

EXPERIMENT 9

Repeated Login Failure Analysis

AIM

Analyze a simulated authentication log, group failed attempts by username and IP address, and report groups exceeding a configurable threshold.

ALGORITHM

1. Start.
2. Read/accept the required evidence data.
3. Process and analyze the data according to the experiment.
4. Apply the required threshold/rules/comparison.
5. Display the findings.
6. Stop.

CODE

```
from collections import defaultdict
log=[("09:00","alice","192.168.1.10","FAIL"),("09:02","alice","192.168.1.10","FAIL"),("09:04","alice","192.168.1.10","FAIL"),("09:06","alice","192.168.1.10","FAIL"),("09:10","bob","192.168.1.20","FAIL")]
threshold=3
g=defaultdict(list)
for t,u,ip,r in log:
    if r=="FAIL": g[(u,ip)].append(t)
for (u,ip),v in g.items():
    if len(v)>=threshold: print(u,ip,"earliest",v[0],"latest",v[-1],"failures",len(v))
```

OUTPUT



The screenshot shows a VS Code editor with a Python script named `EXP-9.py` and its output in the terminal. The script defines a list of login attempts and a threshold of 3. It uses a defaultdict to group failures by username and IP address. The output in the terminal shows that 'alice' at '192.168.1.10' has 4 failures, with the earliest at '09:00' and the latest at '09:06'.

```
1 from collections import defaultdict
2 log=[("09:00","alice","192.168.1.10","FAIL"),("09:02","alice","192.168.1.10","FAIL"),("09:04","alice","192.168.1.10","FAIL"),("09:06","alice","192.168.1.10","FAIL"),("09:10","bob","192.168.1.20","FAIL")]
3 threshold=3
4 g=defaultdict(list)
5 for t,u,ip,r in log:
6     if r=="FAIL": g[(u,ip)].append(t)
7 for (u,ip),v in g.items():
8     if len(v)>=threshold: print(u,ip,"earliest",v[0],"latest",v[-1],"failures",len(v))
9
10
```

Terminal Output:

```
PS C:\Users\sach\AppData\Local\Programs\Microsoft VS Code> & C:\Users\sach\AppData\Local\Python\pythoncore-3.14-64\python.exe "C:\Users\sach\OneDrive\Desktop\digital lab\EXP-9.py"
alice 192.168.1.10 earliest 09:00 latest 09:06 failures 4
PS C:\Users\sach\AppData\Local\Programs\Microsoft VS Code>
```

RESULT

Repeated login failures exceeded the threshold and were reported for investigation.